

Machine Learning Reveals Welfare Discourse Embedded in Japanese Sentencing

A Text Analysis of 1,954 Theft-Related Judgments

Masahiro Sadamura¹ (presenter) · Tomoya Mukai² · Eiichiro Watamura¹

¹ Graduate School of Human Sciences, The University of Osaka ² Fukuyama University

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sadamura.jp/en

Background: sentencing without a PSI system



- Sentencing is a critical juncture in criminal justice — yet large-scale empirical analysis of judicial reasoning remains limited
- **Japan has no pre-sentence investigation (PSI) system for adult criminal courts**
 - Information about defendants' welfare needs reaches judges through ad hoc advocacy, not systematic assessment
- How welfare-related information functions within sentencing reasoning has not been examined at scale

This study



- To our knowledge, the first published NLP / machine-learning study of Japanese criminal sentencing
- Corpus: 1,954 unique theft-related judgments retrieved from Westlaw Japan (48 duplicate retrievals removed)
- Strategy: predict custodial vs. suspended sentence from judgment text, then read the model with explainable-AI tools
 - The model is an instrument for discourse analysis, not a sentencing tool

Data provenance: what the database actually returned



- We set out to collect first-instance district-court judgments, 2010–2023
- The audit of what Westlaw returned: judgments spanning 1958–2026 — only about three-fifths fall within the nominal window
 - Residue of appellate, summary- and family-court decisions (about 3.5% — 69 cases — of the analysis set is appellate)
 - Theft as principal offence: about 1 in 6 cases — most involve mixed offence profiles
- **The database serves up cases selected for legal noteworthiness — not a date- or instance-bounded sample of routine practice**



- Tokenization: MeCab (Japanese morphological analyzer)
- Representation: TF-IDF (5,000 features) + one structured field: prior-conviction count
- Classifier: L1-regularized logistic regression, 10-fold cross-validation
- Explainability: SHAP values on the fitted model
- **Pre-defined lexicon of 59 welfare-related terms, evaluated against the SHAP importance ranking**
- Post-hoc audit (Sept 2026): an 'age' field used in earlier versions mostly recorded other people's ages (victims, witnesses) in these anonymized judgments — withdrawn; all linear results are re-derived without it (text-only AUC 0.901; with prior convictions 0.925)

Result 1: sentencing language is highly patterned



- AUC = 0.925 (text-only 0.901) — highly consistent textual patterns in sentencing
- Strongest predictor: prior-conviction count ($|\text{SHAP}| = 0.331$, toward custodial)
 - Consistent with the recidivism-enhancement principle — the model recovers the shape of sentencing law without being told any law

Result 2: welfare terms sit on the decision boundary



- Of 59 pre-defined welfare-support terms, 14 appear in the top 10% of feature importance
- **That is 2.4 times the rate expected by chance**
 - Robust to a frequency-matched null (observed/expected 2.4–2.5×, empirical $p \leq .004$, $B = 1,000$; in every fold O/E 2.1–2.8×, $p \leq .005$)
- Welfare vocabulary is not peripheral color — it is part of what separates custody from suspension

Result 3: welfare language points both ways



Direction = sign of the fitted weight (of the 26 lexicon terms the L1 model selects, 17 point toward suspension, 9 toward custody; 33 unselected):

- **Toward suspension: e.g., "rehabilitation" (kōsei), "probation" (hogo kansatsu), "supervision" (kantoku)**
- **Toward custody: e.g., "social reintegration" (shakai fukki), "facility" (shisetsu) — plausibly in statements of what is not available; term-level readings are exploratory**
- **Judges reach for the language of welfare both to release and to confine**

Robustness: a second, independent reading



- Fine-tuned Japanese BERT classifier read with integrated gradients (AUC = 0.947)
- **The same welfare vocabulary surfaces under a completely different representation**
 - The pattern is not an artefact of the bag-of-words model
 - Judgment-level cluster bootstrap: $|IG|$ ratio intervals exclude parity in all three audited folds (fold ratios 1.50–2.36 $\times$; pooled 1.85 $\times$, 95% CI [1.60, 2.10])

Interpretation & implications

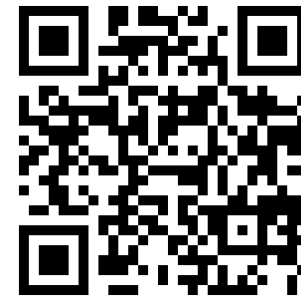


- **Judges already reference welfare concepts in sentencing reasoning — even absent a formal PSI system**
- Institutionalizing welfare-needs assessment would formalize what is already an implicit component of judicial decision-making
- Approach is transferable to other jurisdictions seeking data-driven sentencing analysis
- Caveat: curated, noteworthiness-selected corpus = the world of published contested cases, not routine practice

Thank you — key numbers & contact

- AUC 0.901 text-only / 0.925 with prior convictions / 0.947 BERT · prior convictions |SHAP| = 0.331 · welfare terms 14/59 in top decile (2.4×) · direction 17 suspension / 9 custody (26 selected)
- Enrichment robust to frequency-matched null: O/E 2.4–2.5× ($p \leq .004$); every fold 2.1–2.8× ($p \leq .005$)
- Provenance: 1958–2026 (~3/5 in nominal window) · appellate residue 3.5% · theft-as-principal ≈ 1 in 6
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